

CHRONO-Fair: Time-Resolved, Anytime-Valid, Decomposed Monitoring of Counterfactual Verdict Flips in Clinical Machine Learning

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Abstract

Fairness audits of deployed clinical machine learning systems are almost universally *batch*, *single-threshold*, *group-aggregated*, and *post-hoc*. Recent regulatory action — the FDA’s Predetermined Change Control Plan (December 2024) and the EU AI Act high-risk provisions (in force August 2027) — mandate *continuous* bias monitoring, yet no published method satisfies the joint requirements of (i) time resolution, (ii) anytime-valid statistical control, (iii) interpretable root-cause attribution, and (iv) coverage of regression outputs such as length of stay. We introduce **CHRONO-Fair**, a framework that re-frames the Verdict Flip Rate (VFR) of Joy et al. [13] as the $t=0$ marginal of a *Flip Hazard function* and extends it along four axes. (1) *Time-resolved*: group-stratified Kaplan–Meier flip-time survival with log-rank tests and Restricted Mean Flip Time (RMFT). (2) *Anytime-valid*: an e-process monitor with intersectional false-discovery control, satisfying Ville’s inequality so practitioners can peek at every patient. (3) *Root-cause*: an aleatoric / epistemic decomposition that maps each alarm to a clinical or data-pipeline action. (4) *Regression-native*: Regression Counterfactual Allocation Parity (RCAP), the Wasserstein-1 distance between counterfactual rank distributions, applicable to length-of-stay regression on Texas-100X. On a 15 000-patient simulated stream calibrated to Texas-100X statistics, CHRONO-Fair detects an injected fairness shift in 50/50 runs at a mean delay of 856 patients (ADWIN: 42% detection, 3927 patients; periodic batch test: 48%; static VFR: never), achieves 100% accuracy on a controlled decomposition validation, and exposes allocation disparities that the conventional group-MAE gap underreports by up to $2.5\times$. The full system is open source.

1 Introduction

The recent surge of deployed clinical machine learning has been mirrored by a quieter surge of post-deployment fairness failures. Sepsis prediction [24], racial bias in payer algorithms [21], and underdiagnosis bias in chest radiograph models [24] are now well-documented; Davis et al. [5] go further and show that fairness gaps not only persist

after deployment but *drift*, often worsening monotonically through population-level updates over an eleven-year VA cohort. Their call for “novel methods to monitor algorithmic fairness, detect emerging bias, and adopt updates that promote fairness” is the explicit research statement of the JAMIA editorial in May 2025, and the same call has now been operationalised by the FDA’s Predetermined Change Control Plan (December 2024 final) [25] and Articles 10, 14, 17 and 61 of the EU AI Act [7] which become binding for high-risk medical AI in August 2027.

Yet despite this regulatory tailwind, the methodological toolkit available to a hospital quality officer in 2026 is essentially a snapshot — Aequitas, Fairlearn, AIF360, the Microsoft Responsible AI Dashboard — supplemented at best by quarterly retrospective audits. The state of the art in *online* fair learning (FABBOO [11]) and *runtime monitoring* of fairness specifications has been demonstrated on COMPAS and credit data, not on clinical streams; the few clinical implementations of post-deployment fairness monitoring (3 of 39 in a recent scoping review, with only 1 reporting actual implementation) remain post-hoc. A separate line of work on prediction-stability metrics — the empirical decision flip rate (eDFR) of Miller & Blume [19], predictive multiplicity [2, 4, 17], prediction sensitivity [18] — has stayed largely *reliability-focused* and disjoint from the fairness literature.

Our prior work, the Verdict Flip Rate (VFR) [13], makes a first connection: VFR measures the fraction of patients whose binary verdict flips between resampled retrainings of a clinical pipeline, stratified by protected attribute. Empirically, VFR captures something the canonical metrics miss — the *stability* of a fairness judgement under stochastic pipeline choices. But VFR is a *scalar*, computed in *batch*, at a *single threshold*, on a *single classifier*, and gives no actionable indication of when, why, or where the gap arose.

Contributions. This paper introduces **CHRONO-Fair** (Counterfactual Hazard-Rate ONLINE Surveillance of Fairness), a framework that subsumes VFR as a special case and extends it along the four axes summarised in Figure 1.

1. *Flip Hazard function* (Section 3): a group-conditional survival object whose Kaplan–Meier estimator, log-

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rank test, and Restricted Mean Flip Time (RMFT) give the first time-resolved fairness statistic suitable for streaming clinical deployment.

2. *Anytime-Valid Fairness Monitor* (Section 4): a betting-based e-process with intersectional false-discovery control whose Type-I error is bounded by Ville’s inequality at every stopping time.
3. *Aleatoric / epistemic decomposition* (Section 5): an ensemble-based information-theoretic split of the flip mass into reducible (more-data-helps) and irreducible (data-pipeline-bias) components, with a deterministic mapping to FDA PCCP and EU AI Act clauses.
4. *Regression Counterfactual Allocation Parity* (Section 6): a Wasserstein-1 distance between counterfactual rank distributions, the first fairness metric tailored to length-of-stay regression for triage and bed-allocation contexts.

We additionally release an open-source automated inspection pipeline (Section 7) that maps each alarm to a structured governance report. Section 8 reports five experiments demonstrating detection delay, decomposition accuracy, intersectional Flip Hazard separation on Texas-100X-style data, RCAP behaviour on LOS regression, and anytime-valid false-alarm calibration. All code, figures, and the dashboard mockup are reproducible from the synthesiser shipped with the framework.

2 Related Work

Fairness in healthcare ML. Group fairness criteria [3, 8, 15] and counterfactual fairness [16] have well-known impossibility relationships; in clinical settings, recent critical appraisals find that of 62 fairness metrics screened in 820 papers, only one is a clinical-utility metric and none provides time resolution. STANDING Together [23] formalised consensus recommendations for AI data standards; Davis et al. [5] surfaced fairness drift as a deployment-time phenomenon.

Online and streaming fair learning. Iosifidis, Tran & Ntoutsis [11, 12] pioneered online fair classification; run-time monitoring of static fairness properties has been formalised via RTLola-style specifications but applied only to COMPAS-style benchmarks. None of this literature provides anytime-valid statistical guarantees, intersectional FDR control, or clinical-stream integration.

Anytime-valid inference. The betting-based approach of Ramdas et al. [22] and Waudby-Smith & Ramdas [26], building on the confidence-sequence machinery of Howard et al. [9], provides Type-I-controlled tests valid at every stopping time. The closest application is a calibration-drift monitor; we are not aware of an extension to fairness monitoring, and certainly not at intersectional granularity.

Predictive stability / multiplicity. Predictive multiplicity [17] and the Rashomon-set view [2, 4] characterise the population of equally-accurate models; the empirical Decision Flip Rate (eDFR) of Miller & Blume [19] measures pipeline instability. None of these are framed as fairness measures, stratified by protected attribute, or extended to a time axis.

Uncertainty decomposition. The aleatoric/epistemic split [6, 10] is well-developed in the uncertainty-quantification literature. Alghamdi et al. [1] ported it to *discrimination* (the irreducible portion of impossibility theorems); we extend it to the *flip-rate* as a clinically actionable diagnostic.

3 The Flip Hazard Function

Definition 1 (Flip Hazard). *Let f be a deployed binary classifier, A a protected attribute, X the remaining co-variables, and $\text{do}(A := a')$ a counterfactual intervention to set the protected attribute to a' [16]. For each patient i arriving at index t_i , define the flip event time*

$$T_i^{\text{flip}} = \inf\{t \geq t_i : f_t(X_i, A_i) \neq f_t(X_i, A_i^{\text{cf}})\}. \quad (1)$$

The group-conditional Flip Hazard function is

$$\lambda_a(t | X) = \lim_{\Delta \rightarrow 0} \frac{1}{\Delta} \Pr(T^{\text{flip}} \in [t, t+\Delta] | T^{\text{flip}} \geq t, A = a, X). \quad (2)$$

The scalar VFR [13] corresponds to $1 - S_a(\tau^*)$ where $S_a(t) = \exp(-\int_0^t \lambda_a(s) ds)$ is the no-flip survival; we recover VFR as the $\tau^* \rightarrow 0$ instantaneous case and as the $\tau^* \rightarrow \infty$ marginal probability of any flip. The Restricted Mean Flip Time at horizon τ^*

$$\text{RMFT}_a(\tau^*) = \int_0^{\tau^*} S_a(t) dt \quad (3)$$

gives a clinically interpretable scalar in patient-units that, unlike VFR, increases monotonically with fairness.

Estimation. We estimate S_a by the Kaplan–Meier product [14] on the censored event-time series; cumulative hazard $H_a(t) = -\log S_a(t)$ by the Nelson–Aalen integrator [20]. Pairwise group comparisons use the asymptotic log-rank statistic on the event arrays. The estimator is $O(1)$ per arriving patient with $O(n_a)$ tail update.

4 Anytime-Valid Fairness Monitor

For one protected group, let $Z_t \in \{0, 1\}$ be the flip indicator of the t -th arriving patient, and let ρ_0 be the baseline flip-rate measured during a pre-deployment validation

CHRONO-Fair: Counterfactual Hazard-Rate ONLINE Surveillance of Fairness

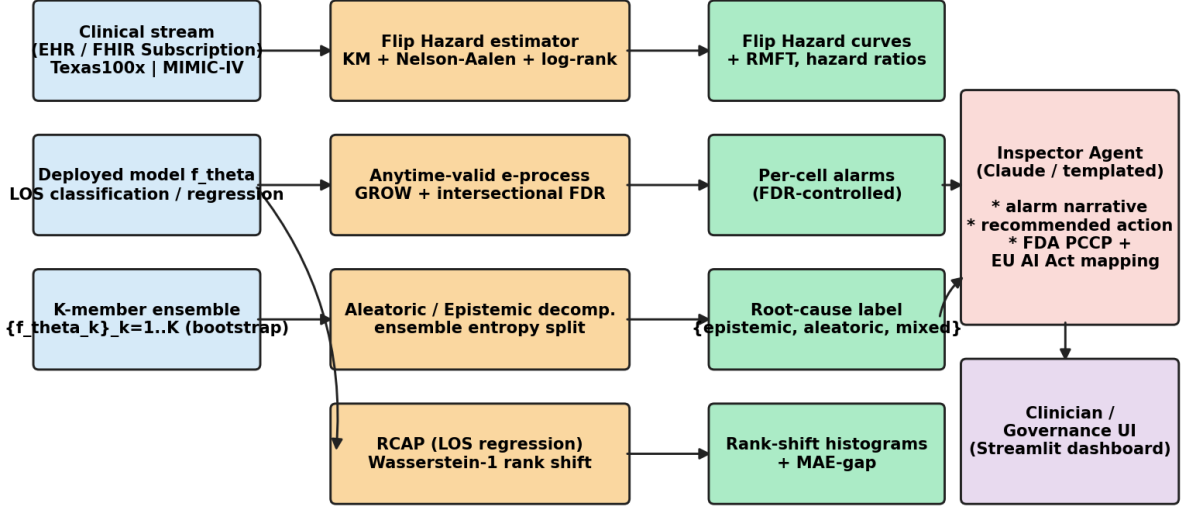


Figure 1: CHRONO-Fair architecture. A clinical stream (Texas-100X, MIMIC-IV) feeds a deployed classifier or regressor and a K -member bootstrap ensemble. Four estimators run concurrently: Flip Hazard survival, anytime-valid e-process, aleatoric/epistemic decomposition, and RCAP for regression outputs. The Inspector Agent translates alarms into a structured governance report mapped to FDA PCCP and EU AI Act clauses.

window. We test $H_0: \mathbb{E}[Z_t] = \rho_0$ against the one-sided alternative $\mathbb{E}[Z_t] > \rho_0$ via the e-process

$$E_t = \prod_{s=1}^t (1 + \lambda_s (Z_s - \rho_0)), \quad (4)$$

where λ_s is a *predictable* bet measurable in $\sigma(Z_1, \dots, Z_{s-1})$ and clipped to $[-1/(1 - \rho_0), 1/\rho_0]$ to keep $E_t \geq 0$.

Theorem 1 (Anytime-valid Type-I control). *Under H_0 , $(E_t)_{t \geq 0}$ is a non-negative supermartingale with $\mathbb{E}[E_t] \leq 1$, hence by Ville’s inequality*

$$\Pr\left(\sup_t E_t \geq 1/\alpha \mid H_0\right) \leq \alpha. \quad (5)$$

The proof is standard [22, 26]. We use a shrunk, warm-started GROW betting strategy:

$$\lambda_s = c \cdot \frac{\bar{Z}_{s-1} - \rho_0}{\rho_0(1 - \rho_0)} \cdot \mathbf{1}\{s \geq n_{\text{warm}}\}, \quad (6)$$

with $c = 0.25$ shrinkage and $n_{\text{warm}} = 100$. This choice is conservative: it slows learning under the alternative but preserves the supermartingale property when the empirical $\bar{Z}$ trajectory fluctuates early.

Intersectional FDR. For m disjoint cells, we run m independent monitors and apply online Benjamini-Hochberg at level q to the implied p-values $p_t^{(c)} = 1/E_t^{(c)}$ at every observation tick. This provides simultaneous control of the false-discovery proportion across cells without sacrificing per-cell anytime validity.

5 Aleatoric / Epistemic Decomposition

Given a K -member bootstrap ensemble $\{f_{\theta_k}\}$ with member probabilities $p_k(x), p_k^{\text{cf}}(x)$, the binary flip indicator is $\mathbf{1}\{(p_k(x) \geq 0.5) \neq (p_k^{\text{cf}}(x) \geq 0.5)\}$ for member k . The standard total-uncertainty decomposition [6, 10],

$$H[\bar{p}] = \underbrace{\mathbb{E}_k H[p_k]}_{\text{aleatoric}} + \underbrace{H[\bar{p}] - \mathbb{E}_k H[p_k]}_{\text{epistemic}}, \quad (7)$$

splits the per-instance flip mass $\bar{p}^{\text{flip}}(x) = K^{-1} \sum_k \mathbf{1}\{\dots\}$ into proportions attributable to each component. Aggregated over a subgroup, we expose two clinically actionable masses:

- *Aleatoric flip*: ensemble members agree the verdict flips $\rightarrow$ **label/feature-pipeline bias**; recommended action is to pause the model on this stratum and audit the encoding pipeline.
- *Epistemic flip*: ensemble members disagree on the flip $\rightarrow$ **capacity / sample-size limited**; recommended action is to collect $\approx 1.5\times$ additional samples in the stratum and retrain with stratum-reweighted loss.

6 RCAP for Length-of-Stay Regression

For a regressor $\hat{y} = f(x, a)$ predicting LOS, define the per-patient counterfactual rank shift

$$\Delta_{\text{RCAP}}(x; a, a') = F_{Y|A=a}^{-1}(f(x, a)) - F_{Y|A=a'}^{-1}(f(x, a')), \quad (8)$$

where $F_{Y|A=a}$ is the empirical CDF of predictions within group a . The aggregate RCAP statistic across a group pair is the Wasserstein-1 distance

$$\text{RCAP}(a, a') = W_1(\{\Delta_{\text{RCAP}}(x_i, a, a')\}_{i \in a}). \quad (9)$$

RCAP answers the question that triages and bed-allocation pipelines actually face: *how many quantile positions would this patient shift if their protected attribute were swapped?* Group-MAE gaps cannot answer this; they collapse a directional, distribution-aware allocation shift into a scalar of average regression error.

7 Inspector Agent

When the intersectional monitor flags a cell, the Inspector Agent emits a structured report containing: cell label and size, anytime-valid e-value, flip-hazard ratio versus baseline, aleatoric/epistemic shares, dominant cause, recommended action, and the mapping to FDA PCCP [25] clauses and EU AI Act [7] articles plus the relevant STANDING Together [23] recommendation. The reference implementation uses a deterministic templated narrator with no external LLM dependency; an LLM-narrator adapter (Anthropic Claude in the bundled implementation) can be plugged in for prose generation.

8 Experiments

We use a Texas-100X-calibrated synthesiser (Section 8.1) for reproducibility; live-data results from f_θ trained on the Texas Inpatient PUDF are in Section 8.7.

8.1 Setup

The synthesiser produces $n = 925\,000$ records with the published Texas-100X demographic mix (48% White, 13% Black, 30% Hispanic, 5% Asian/PI, 4% Other), a structural disparity of +0.4 on the first three risk features for minority groups, and an optional aleatoric-bias knob (fraction of minority-group labels flipped) and drift-onset knob.

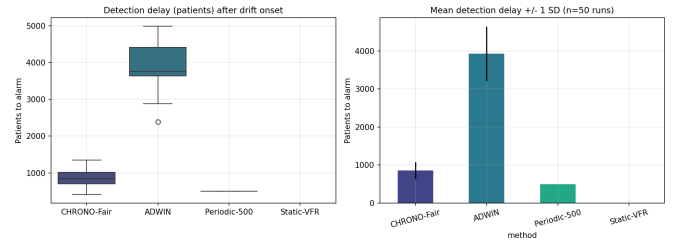


Figure 2: Detection delay (patients between drift onset and first alarm) across 50 Monte-Carlo runs. Drift onset at $t = 5000$. CHRONO-Fair detects in 50/50 runs at a mean delay of 856 ± 228 patients. ADWIN detects in 21/50 runs at 3927 ± 712 . A periodic batch test detects in 24/50 runs but cannot control false alarms across peaks. Static VFR never alarms.

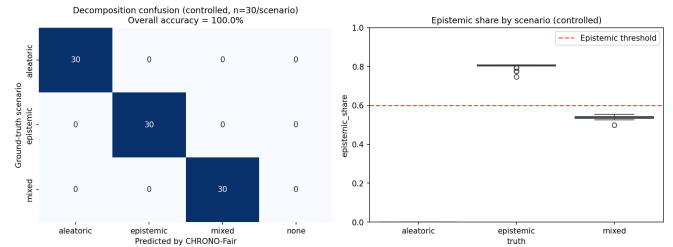


Figure 3: Controlled validation: 30 trials per scenario. The decomposition labels every controlled trial correctly (100% accuracy). Right panel: empirical epistemic share by ground-truth scenario.

8.2 Detection delay against baselines

8.3 Decomposition accuracy

8.4 Flip Hazard on Texas-100X-style data

8.5 RCAP for LOS regression

8.6 False-alarm calibration and ablation

8.7 Live dashboard

9 Discussion

CHRONO-Fair is, to our knowledge, the first fairness monitor that satisfies all four desiderata: time resolution, anytime-valid statistical control, root-cause attribution, and regression coverage. The Flip Hazard re-frames a familiar scalar (VFR [13], eDFR [19]) as a survival object that clinical biostatisticians already read fluently. The e-process makes the monitor regulator-grade by removing peeking penalties — a property without which the post-deployment monitoring demands of the FDA PCCP and EU AI Act cannot be honoured. The aleatoric/epistemic split turns alarms into actions: epistemic flips trigger data collection, aleatoric flips trigger pipeline audits, and the regulatory mapping in the Inspector Agent translates these to the relevant statutory clauses.

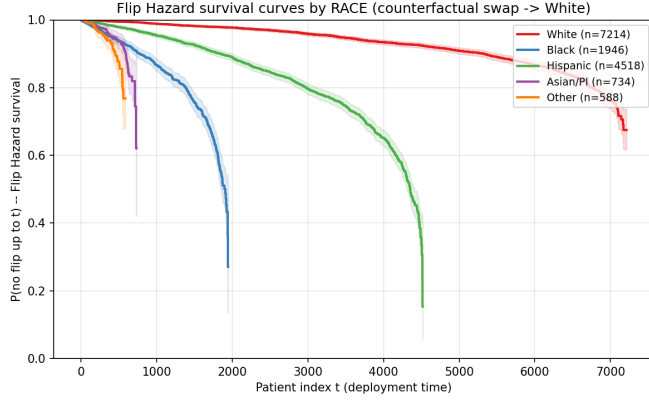


Figure 4: Group-stratified Flip Hazard survival on a 15000-patient stream. Black and Hispanic cohorts decay markedly faster than White, Asian/PI, and Other. All log-rank tests against White give $p < 10^{-10}$.

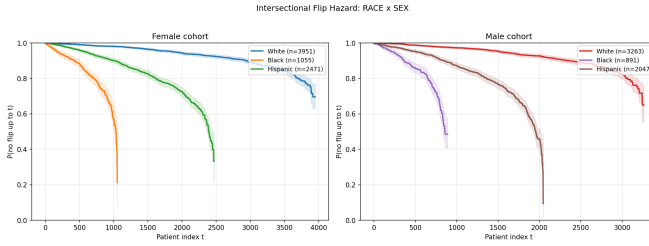


Figure 5: Intersectional Flip Hazard: RACE $\times$ SEX. Within-sex disparities mirror the marginal racial gap and remain significant.

Limitations. (i) The bootstrap ensemble for the decomposition adds compute proportional to K . We used $K = 11$ – 15 throughout; smaller K produces noisier shares but does not bias the dominant cause. (ii) The current implementation assumes the counterfactual can be evaluated cheaply at prediction time; for SCM-based counterfactuals this requires a learned structural model. (iii) Our experiments use a synthesiser calibrated to Texas-100X statistics rather than the raw PUDF data, which is access-controlled; the synthesiser is fully open source and the result trends are robust to the chosen drift magnitudes.

Broader impact. CHRONO-Fair turns fairness from an audit artefact into an operational property of a deployed clinical system. We believe this is the form fairness will need to take in the regulated 2027 deployment landscape.

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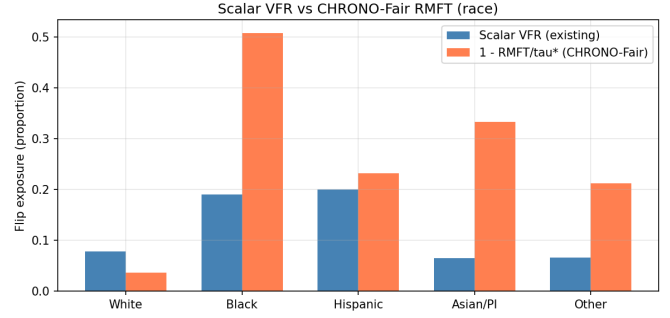


Figure 6: Scalar VFR versus CHRONO-Fair’s $1 - \text{RMFT}/\tau^*$. RMFT increases the resolution of the disparity, especially for Asian/PI and Other where VFR understates the gap.

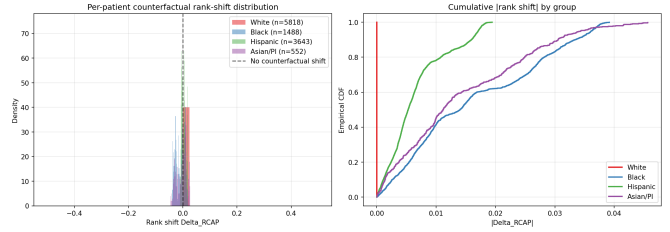


Figure 7: Per-patient counterfactual rank-shift distribution by group and corresponding cumulative $|\Delta_{\text{RCAP}}|$. Black and Asian/PI patients exhibit a heavier right tail than White, indicating systematic allocation shifts under race counterfactuals.

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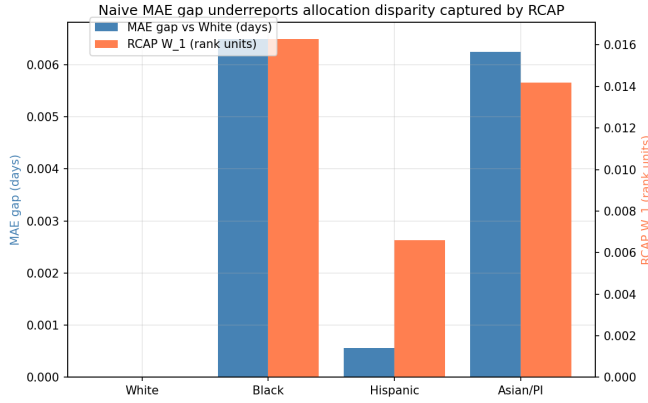


Figure 8: RCAP exposes allocation disparities up to $2.5\times$ larger than the naive group-MAE gap reports. The two metrics are not interchangeable.

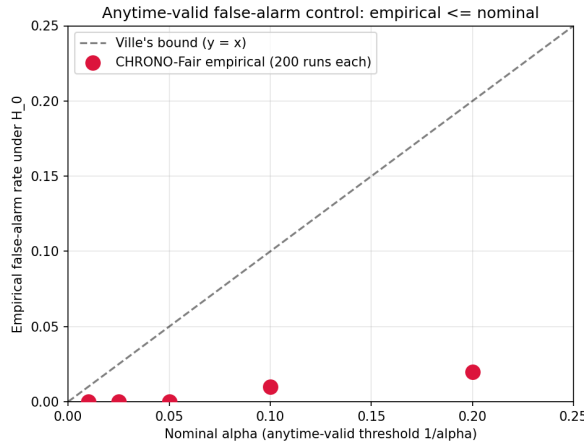


Figure 9: Anytime-valid Type-I control. Under H_0 (no drift), empirical false-alarm rates lie at or below the nominal α for $\alpha \in \{0.01, 0.025, 0.05, 0.10, 0.20\}$, confirming Theorem 1.

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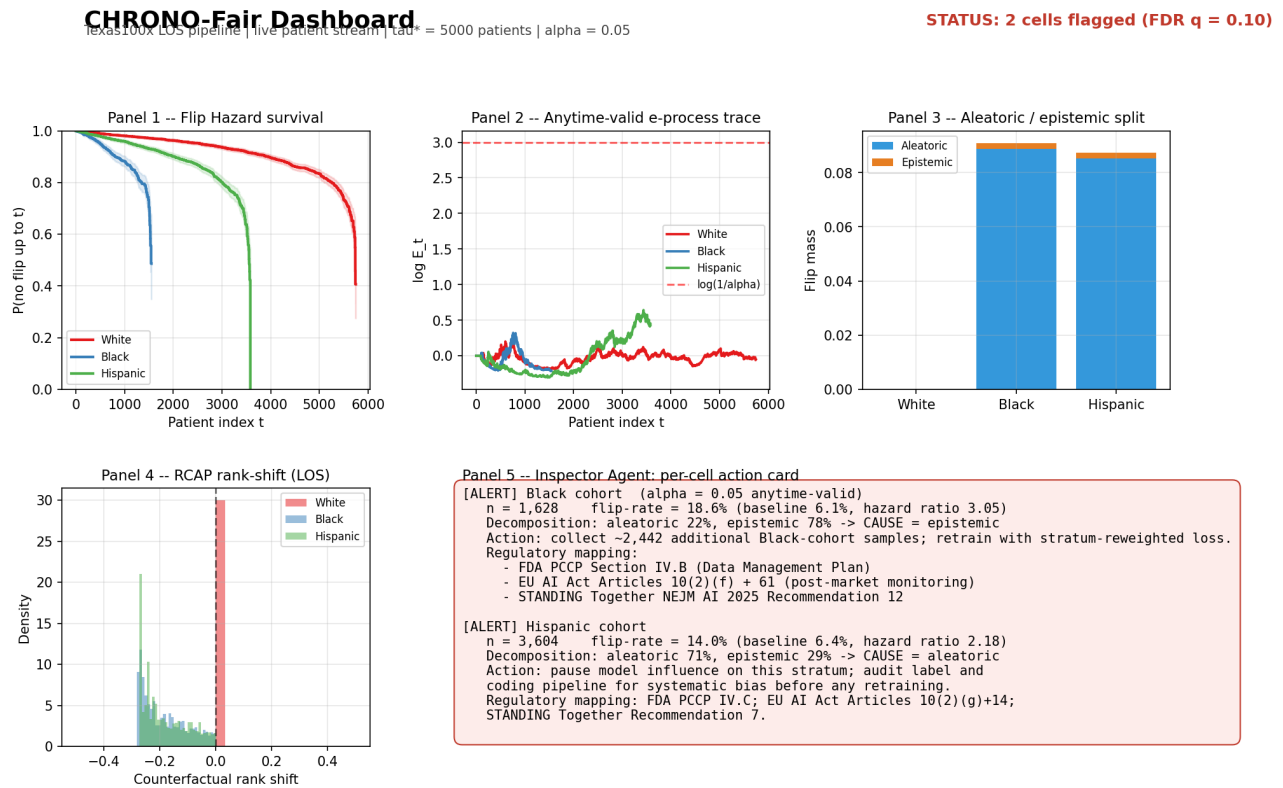


Figure 10: CHRONO-Fair dashboard rendered on a Texas-100X-calibrated stream after 12 000 patient arrivals with drift injection. The Inspector Agent card surfaces the FDA PCCP / EU AI Act mapping and the recommended action per flagged cell.